



Activity File - Exploiting Vulnerable Templates

In this activity, you will learn how attackers can forge certificate requests to elevate privileges by exploiting misconfigured certificate templates in Active Directory Certificate Services (AD CS). Follow the steps below to understand the process.

1. Review the misconfigured template.

- Start by reviewing the certificate template identified as misconfigured in the previous activity.
- Ensure you understand the specific vulnerabilities, such as overly permissive enrollment permissions or weak cryptographic settings.

2. Set up the attack environment.

- On your attack machine (e.g., Kali Linux or Windows machine with administrative privileges), ensure you have the necessary tools installed, such as Certify or Rubeus.
- If using Windows, download the Certify tool from GitHub. If you're using Kali, ensure you have Rubeus configured.

3. Forge a certificate request with Certify.

- On your Windows attack machine, open a command prompt as an administrator.
- Use Certify to enumerate available certificate templates by running the following command:

Unset

```
Certify.exe find /vulnerable
```

- Identify the vulnerable template you discovered earlier. Once identified, forge a certificate request using the following command:

Unset

```
Certify.exe request /ca:<CAName> /template:<TemplateName>  
/altname:<Domain\Username>
```

- Replace `<CAName>`, `<TemplateName>`, and `<Domain\Username>` with the actual names based on your environment.

4. Obtain the certificate.

- After running the command, a certificate will be generated. Save this certificate to a file on your machine.
- Verify that the certificate has been successfully created and contains the information needed to impersonate the target user or service.

5. Use the certificate for privilege escalation.

- With the obtained certificate, use tools like Mimikatz or Rubeus to impersonate the user associated with the certificate.
- Run the following Rubeus command to pass the certificate and gain access as the privileged user:

Unset

```
Rubeus.exe asktgt /user:<Username>  
/certificate:<PathToCertificate> /password:<Password>
```

- Replace `<Username>`, `<PathToCertificate>`, and `<Password>` with the appropriate values.

6. Document your findings.

- Record the steps you took to forge and use the certificate, noting any challenges or observations.
- Take screenshots of the successful certificate request and the resulting elevated access.